

Advanced Python — Hard 2-Mark Questions

Loop Manipulation • Predict the Output

Instruction: Predict the exact output of each Python program.

Q1. Predict the output. [2 Marks]

```
x = 1

for i in range(2, 6):
    for j in range(i, 0, -1):
        if (i + j) % 2 == 0:
            x += j
        else:
            x -= i

    if x > 4:
        break

print(x, i, j)
```

Q2. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 6):
    for j in range(1, i + 1):
        if j == 3:
            continue
        x += i - j

    if x >= 5:
        break

print(x, i, j)
```

Q3. Predict the output. [2 Marks]

```
x = 1

for i in range(2, 6):
    for j in range(i, 0, -1):
        if x % 2 == 0:
            x += j
        else:
            x -= 1

    if x < 0:
        break

print(x, i, j)
```

Q4. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 5):
    for j in range(i):
        if j == 1 and i % 2 == 0:
            break
        x += i + j

    if x > 10:
        break

print(x, i, j)
```

Q5. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 6):
    for j in range(1, 4):
        if i == j:
            continue
        x += j

    if x > 8:
        break

print(x, i, j)
```

Q6. Predict the output. [2 Marks]

```
x = 10

for i in range(1, 5):
    for j in range(i):
        if j % 2:
            x -= i
        else:
            x += j

    if x < 8:
        break

print(x, i, j)
```

Q7. Predict the output. [2 Marks]

```
x = 0

for i in range(2, 7):
    for j in range(1, i):
        if (i * j) % 3 == 0:
            x += 1
        else:
            x -= 1

    if x == 0:
        break

print(x, i, j)
```

Q8. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 6):
    for j in range(i):
        if j == 0:
            x += i
        elif x % 2 == 0:
            continue
        else:
            x -= j

    if x > 7:
        break

print(x, i, j)
```

Q9. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 6):
    for j in range(1, i + 1):
        x += (-1) ** j * i

    if x < -2:
        break

print(x, i, j)
```

Q10. Predict the output. [2 Marks]

```
x = 0

for i in range(1, 6):
    for j in range(i):
```

```
        if (i + j) % 3 == 0:
            continue
        x += j

    if x >= 6:
        break

print(x, i, j)
```